

Maria Strazzullo

Non-tenure track Assistant Professor at DISMA, Politecnico di Torino

maria.strazzullo@polito.it

Personal Information

Name	Maria Strazzullo
Date of birth	March 9th, 1992
Address	Corso Duca degli Abruzzi, 24, 10129 Torino

Academic Experience

Assistant professor Non-tenure tracked, DISMA, Politecnico di Torino, Turin, Italy	January 2024–ongoing
Postdoc Excellence project fellowship, DISMA, Politecnico di Torino, Turin, Italy	January 2022–January 2024
Fellowship MathLab group, SISSA, Trieste, Italy	October–December 2021
Ph.D. Applied Mathematics mathLab group, SISSA, Trieste, Italy	October 2017–September 2021
Predoc mathLab group, SISSA, Trieste, Italy	April–September 2017

Scientific Interests

Reduced order methods, applied mathematics, optimal control theory, turbulence modelling, uncertainty quantification, environmental and ecological sciences, neural networks for partial differential equations.

Education

International School for Advanced Studies (SISSA), Trieste PhD in Mathematical Analysis, Modelling, and Applications; cum laude. Thesis: “Model Order Reduction for Nonlinear and Time-Dependent Parametric Optimal Flow Control Problems”. Advisor: Prof. Gianluigi Rozza; Co-advisor: Dr. Francesco Ballarin.	September 24, 2021
Università degli Studi di Trieste Master’s degree in Mathematics, 110/110 cum laude. Thesis: “Reduced order methods for parametrized optimal flow control problems”. Advisor: Prof. Gianluigi Rozza; Co-advisors: Prof. Renzo Mosetti, Dr. Francesco Ballarin.	Sep 2014–Mar 2017
Università degli Studi di Camerino Bachelor’s degree in Mathematics, 110/110 cum laude. Thesis: “La Teoria Dei Codici Autocorrettori” (“The theory of error-correcting codes”). Advisor: Prof. Carlo Toffalori.	Sep 2011–Jul 2014
Liceo Classico “Giacomo Leopardi”, Macerata Classical certificate, 100/100.	Sep 2006–Jul 2011

Visits and Fellowships

Research in pairs fellowship With Dr. Luca Saluzzi, Centro Ennio De Giorgi, Pisa, Italy	September 6–13, 2024
Visiting invited researcher University of Siviglia	June–July 2024; July–August 2023
Visiting invited researcher Virginia Tech, Interdisciplinary Center for Applied Mathematics (ICAM)	October–November 2023; October–November 2022

Awards and Grants

- 2026 SIMAI Prize for Young Researchers in Industrial Mathematics.
- ASN – abilitazione scientifica nazionale.
- ECCOMAS Young Investigator Grant 2024: “CRAFT: Control and Reg-reduction in Applications for Flow Turbulence” (€15,000).
- PI, INdAM–GNCS Project CUP_E53C24001950001: “Metodi numerici efficienti per problemi accoppiati in sistemi complessi”, 2025 (€2,800).
- Travel funds for EUCCO 2023 (€380); INdAM fellowship for a month abroad, 2023 (€3,000 plus travel funds).

- PI, INDAM–GNCS Project CUP_E53C22001930001: “Metodi numerici per lo studio di strutture geometriche parametriche complesse”, 2023 (€4,300).
- GNCS grant for ECCOMAS 2022 (€1,000); BGCE Prize finalist and SIAM Student Travel Award, SIAM-CSE 2021; ECCOMAS Scholarship for WCCM-ECCOMAS 2021; Special Mention, PhD4Innovating, ESOF 2020.
- Participant, 2024 INDAM–GNCS project “Metodi di riduzione di modello e approssimazioni di rango basso per problemi alto-dimensionali” (€4,800); MIT–Friuli Venezia Giulia Seed Fund 2019–2020; 2018 INDAM–GNCS Project “Model Reduction in Medical Applications”.

Publications and Preprints

- [29] **Preprint** K. Cascavita, F. Marcon and M. Strazzullo, “Numerical Analysis of the Virtual Element Approximation for the Smagorinsky Turbulence Model”, 2026. <https://arxiv.org/abs/2609.02433>.
- [28] **Preprint** F. Della Santa, S. Pieraccini and M. Strazzullo, “Graph-Instructed Neural Networks for parametric problems with varying boundary conditions”, 2026. <https://arxiv.org/abs/2603.08304>.
- [27] **Preprint** A. Ivagnes, M. Strazzullo and G. Rozza, “Reinforcement Learning-Based Filters for Convection-Dominated Flows: Reference-Free and Reference-Guided Training”, 2026. <https://arxiv.org/abs/2603.02086>.
- [26] **Book (Editor)** F. Pichi, G. Rozza, M. Strazzullo and D. Torlo (eds.), *Scientific Machine Learning Emerging Topics*, 2026. <https://link.springer.com/book/9783032115263>.
- [25] **Paper** D. Oberto, M. Strazzullo and S. Berrone, “Machine Learning enhanced parametric Reynolds-averaged Navier-Stokes equations at the full and reduced order levels”, IJNME, 2026. <https://arxiv.org/abs/2510.06992>.
- [24] **Paper** S. Berrone et al., “General order virtual element approximation for the Smagorinsky turbulence model”, Mathematics and Computers in Simulation, 2026. <https://doi.org/10.1016/j.matcom.2026.06.009>.
- [23] **Preprint** A. Borio, F. Marcon and M. Strazzullo, “Stabilization-Free General Order Virtual Element Methods for Neumann Boundary Optimal Control Problems in Saddle Point Formulation”, 2024. <https://arxiv.org/abs/2411.08497>.
- [22] **Preprint** M. Strazzullo, F. Ballarin, T. Iliescu and C. Canuto, “New Feedback Control and Adaptive Evolve-Filter-Relax Regularization for the Navier-Stokes Equations in the Convection-Dominated Regime”, 2023. <https://arxiv.org/abs/2307.00675>.
- [21] **Paper** L. Saluzzi and M. Strazzullo, “Dynamical Low-Rank Approximation Strategies for Nonlinear Feedback Control problems”, Journal of Numerical Mathematics, 2025. <https://dx.doi.org/10.1515/jnma-2025-0005>.
- [20] **Paper** F. Pichi and M. Strazzullo, “Deflation-based certified greedy algorithm and adaptivity for bifurcating nonlinear PDEs”, CNSNS, 2025. <https://doi.org/10.1016/j.cnsns.2025.108941>.
- [19] **Paper** A. Ivagnes et al., “Data-driven Optimization for the Evolve-Filter-Relax regularization of convection-dominated flows”, IJNME, 2025. <https://onlinelibrary.wiley.com/doi/10.1002/nme.70042>.
- [18] **Paper** M. Strazzullo et al., “Variational Multiscale Evolve and Filter Strategies for Convection-Dominated Flows”, CMAME, 2025. <https://doi.org/10.1016/j.cma.2025.117811>.
- [17] **Paper** M. Strazzullo and F. Vicini, “POD-based reduced order methods for optimal control problems governed by parametric PDE with varying boundary control”, AMC, 2023. <https://doi.org/10.1016/j.amc.2023.128191>.
- [16] **Paper** F. Zoccolan, M. Strazzullo and G. Rozza, “Stabilized Weighted Reduced Order Methods for Parametrized Advection-Dominated Optimal Control Problems with Random Inputs”, JNM. <https://doi.org/10.1515/jnma-2023-0006>.
- [15] **Paper** F. Zoccolan, M. Strazzullo and G. Rozza, “A Streamline upwind Petrov-Galerkin Reduced Order Method for Advection-Dominated PDEs under Optimal Control”, CMAM, 2024. <https://doi.org/10.1515/cmam-2023-0171>.
- [14–12] **Chapters** D. Torlo et al.; M. Strazzullo et al.; F. Pichi et al., chapters in *Advanced Reduced Order Methods and Applications in Computational Fluid Dynamics*. <https://doi.org/10.1137/1.9781611977257>.
- [11] **Proceeding** E. Donadini et al., “A data-driven partitioned approach for time-dependent optimal control problems with dynamic mode decomposition”, ICOSAHOM, 2022. https://link.springer.com/chapter/10.1007/978-3-031-20432-6_13.
- [10] **Paper** N. Demo, M. Strazzullo and G. Rozza, “An Extended Physics Informed Neural Network For Preliminary Analysis of Parametric Optimal Control Problems”, C&MA, 2023. <https://doi.org/10.1016/j.camwa.2023.05.004>.
- [9] **Paper** M. Strazzullo et al., “Consistency of the Full and Reduced Order Models for Evolve-Filter-Relax Regularization”, IJNME, 2022. <https://doi.org/10.1002/nme.6942>.
- [8] **Preprint** M. Strazzullo, F. Ballarin and G. Rozza, “A Certified Reduced Basis Method for Linear Parametrized Parabolic Optimal Control Problems in Space-Time Formulation”, 2021. <https://arxiv.org/abs/2103.00460>.
- [7] **Paper** G. Carere et al., “Weighted POD-reduction for parametrized PDE-constrained Optimal Control Problems with random inputs”, C&MA 102, 261–276, 2021. <https://doi.org/10.1016/j.camwa.2021.10.020>.
- [6] **Chapter** F. Ballarin, G. Rozza and M. Strazzullo, “Space-time POD-Galerkin approach for parametric flow control”, Handbook of Numerical Analysis, 2022. <https://doi.org/10.1016/bs.hna.2021.12.009>.
- [5] **Paper** F. Pichi et al., “Driving bifurcating parametrized nonlinear PDEs by optimal control strategies”, ESAIM: M2AN, 2022. <https://doi.org/10.1051/m2an/2022044>.
- [4] **Paper** M. Strazzullo, F. Ballarin and G. Rozza, “POD-Galerkin Model Order Reduction for Parametrized Nonlinear

Time Dependent Optimal Flow Control”, JNM, 2021. <https://doi.org/10.1515/jnma-2020-0098>.

[3] **Paper** M. Strazzullo, F. Ballarin and G. Rozza, “POD-Galerkin Model Order Reduction for Parametrized Time Dependent Linear Quadratic Optimal Control Problems”, JSC 83(3), 55, 2020. <https://doi.org/10.1007/s10915-020-01232-x>.

[2] **Proceeding** M. Strazzullo et al., “Reduced order methods for parametrized non-linear and time dependent optimal flow control problems”, ENUMATH2019, 2020. https://doi.org/10.1007/978-3-030-55874-1_83.

[1] **Paper** M. Strazzullo et al., “Model Reduction for Parametrized Optimal Control Problems in Environmental Marine Sciences and Engineering”, SIAM J. Sci. Comput. 40(4), B1055–B1079, 2018. <https://doi.org/10.1137/17M1150591>.

Complete list of Talks at Conferences and Seminars

MORE 2026, Milan	Nov 2–5, 2026
Plenary speaker	TBA
INDAM StructApp, Rome	Sep 7–11, 2026
Invited speaker	TBA
WCCM–ECCOMAS, Munich	Jul 19–24, 2026
Plenary speaker	Reduced Order Models for Regularized Convection-Dominated Flows
Young-SIMAI, Pisa	Jun 3–5, 2026
Young plenary speaker	Enhancing Spatial Filters for Convection-Dominated Flows
SPARCL, Paris	Apr 8–10, 2026
Invited speaker	Reinforcement Learning for Filter-Based Regularization
GNCS, Rimini	Feb 3–5, 2026
Invited speaker	Metodi numerici efficienti per problemi accoppiati in sistemi complessi
MATH4AIML, Rome	Jan 21–23, 2026
Contributed talk	Graph-Instructed Neural Network for Varying Boundary PDEs
EUCCO, Klagenfurt	Sep 29–Oct 2, 2025
Invited minisymposium speaker	Dynamical Low-Rank Approximation for Nonlinear Feedback Control
Optimal Control and Agent Systems, Rome	Sep 22–25, 2025
Invited speaker	Dynamical Low-Rank Approximation for Nonlinear Feedback Control
ECCOMAS YIC, Pescara	Sep 17–19, 2025
Plenary speaker	Regularized Reduced Order Models for Convection-Dominated Flows
SIMAI, Trieste	Sep 1–5, 2025
Invited minisymposium speaker	Reinforcement Learning for Filter-Based Regularization
ADMOS, Barcelona	Jun 9–11, 2025
Invited minisymposium speaker	VMS-Enhanced Approaches to Evolve-Filter Strategies
COUPLED, Villasimius	May 25–28, 2025
Invited minisymposium speaker	Dynamical Low-Rank Approximation for Parametric Nonlinear Feedback Control
EMS-TAG-SCIML, Milan	Mar 24–26, 2025
Invited speaker	Machine Learning for Turbulence Modeling
ARIA final workshop, Trieste	Nov 25–27, 2024
Invited speaker	Enhancing spatial filters: VMS-filters and parameter optimization
Sapienza seminar, Rome	Nov 5, 2024
Invited speaker	Dynamical Low-Rank Approximation for Nonlinear Feedback Control
UniPi seminars, Pisa	Sep 9, 2024
Invited speaker	Nonlinear Feedback Control via Dynamical Low-Rank Approximation
AMS–UMI, Palermo	Jul 23–26, 2024
Invited minisymposium speaker	Dynamical low-rank approximations for parametric infinite horizon optimal control
ECM, Seville	Jul 15–19, 2024
Invited minisymposium speaker	Regularized Feedback Control strategies for Convection-Dominated Navier-Stokes Equations
ECCOMAS, Lisbon	Jun 3–7, 2024
Invited minisymposium speaker	Two talks: deflated-greedy certification of bifurcations; regularized ROMs for convection-dominated flows
New Trends in Optimal Control, Venice	May 14–17, 2024
Invited speaker	Regularized full and reduced feedback control strategies
SISSA Women in Mathematics, Trieste	May 13, 2024
Invited speaker	Regularized Feedback Control strategies
MASCOT-NUM, Hyères	Apr 3–5, 2024
Invited speaker	Reduced Order Methods for Parametric Optimal Control
SIAM UQ, Trieste	Feb 27–Mar 1, 2024

Invited minisymposium speaker	Evolve-Filter-Relax strategies in feedback control	
POD UQ, Trieste		Feb 26, 2024
Invited speaker	Optimization problems governed by PDEs under random variables	
GNCS, Rimini		Feb 14–17, 2024
Invited speaker	Metodi numerici per strutture geometriche parametriche complesse	
EUCCO, Heidelberg		Sep 25–28, 2023
Invited minisymposium speaker	Model order reduction for varying boundary optimal control problems	
ECCOMAS YIC, Oporto		Jun 19–21, 2023
Invited minisymposium speaker	Model order reduction for varying boundary optimal control problems	
Math 2 Product, Taormina		May 30–Jun 1, 2023
Invited minisymposium speaker	POD-Based Strategies For Varying Boundary Optimal Control	
International Workshop on ROM, Singapore		May 22–26, 2023
Invited speaker	Model order reduction for parametric optimal control problems	
CFC Congress, Cannes		Apr 25–28, 2023
Invited minisymposium speaker	Full and Reduced EFR regularization in feedback control	
SIAM CSE, Amsterdam		Feb 26–Mar 3, 2023
Invited minisymposium speaker	Model Order Reduction for Parametric Optimal Control Problems in Space-Time Formulation	
ARIA–VT Seminar, Blacksburg		Feb 2, 2023
Invited speaker	The role of EFR Regularization in Feedback Control	
Politecnico di Torino		Nov 24, 2022
Contributed talk	Physics-informed Neural Networks for PDEs and optimal control	
Virginia Tech Math Colloquium		Nov 17, 2022
Invited speaker	Model order reduction for nonlinear and time-dependent optimal control	
Emory Mathematics Seminars		Oct 17, 2022
Invited speaker	Model order reduction: time-dependency to nonlinearity	
GIMC SIMAI YOUNG, Pavia		Sep 29–30, 2022
Invited minisymposium speaker	Physics-informed Neural Networks for parametric PDEs	
MORE, Berlin		Sep 19–23, 2022
Contributed talk	FOM and ROM Consistency for EFR Regularization	
ECCOMAS, Oslo		Jun 5–9, 2022
Invited minisymposium speaker	Optimal control and bifurcating systems	
FAU Mini-Workshop		May 24, 2022
Invited speaker	Model order reduction for time-dependent optimal control	
SIAM UQ		Apr 12–15, 2022
Invited minisymposium speaker	Stabilized ROMs for Transport Control with Random Inputs	
Analysis Junior Seminar		Feb 18, 2022
Invited speaker	FOM and ROM Consistency for EFR Regularization	
Pitt AWM Student Seminar		Dec 3, 2021
Invited speaker	Optimal control in bifurcating phenomena	
COUPLED		Jun 14–16, 2021
Contributed talk	ROMs for UQ Problems in Environmental Sciences	
FEniCS Conference		Mar 22–26, 2021
Contributed talk	Reduced order methods for optimal flow control	
DISMA Seminar Series		Mar 22, 2021
Invited speaker	A Glimpse of ROMs for Parametrized Optimal Control	
SIAM-CSE		Mar 1–5, 2021
Invited minisymposium speaker	Two talks on space-time and nonlinear optimal flow control	
WCCM–ECCOMAS		Jan 11–15, 2021
Invited minisymposium speaker	ROMs for Optimal Flow Control	
MORSS		Sep 7–10, 2020
Contributed talk	Advances in ROMs for Optimal Flow Control	
SAMM		Jul 19–24, 2020
Poster	POD-Galerkin reduction for nonlinear time-dependent optimal flow control	
Summer School on ROM, SISSA		Jul 8–12, 2019
Lecturer and poster presenter	ROMs for parametrized optimal flow control	
ADMOS, El Campello		May 27–29, 2019
Invited minisymposium speaker	ROMs for nonlinear time-dependent optimal flow control	
Analysis, Control and Inverse Problems, Naples		Nov 26–30, 2018

Invited junior speaker

MoRePas, Nantes

Poster

QUIET, SISSA

Poster

ROMs for Optimal Flow Control

Apr 10–13, 2018

POD-Galerkin ROMs for inverse and multi-physics problems

Jul 18–21, 2017

ROMs for Environmental Marine Problems by Optimal Flow Control

Teaching, Mentorship, Organization and Other Tasks

Teaching

- Lecturer, PhD course “Model order reduction for parametric nonlinear problems”, Politecnico di Torino 2025–2026.
- Support lecturer, “Model Order Reduction and Machine Learning”, 2023–2026; “Metodi e Modelli Numerici”, Politecnico di Torino, 2022–2025.
- Lecturer, Summer School on Reduced Order Methods in CFD, Trieste, 2019 and 2022; support lecturer, SISSA AMMA PhD–MHPC course, 2022; support lecturer, Numerical Analysis, Data Science and Scientific Computing, 2018.

Mentorship

- Advisor, Luigi Donnarumma: “Data-driven local time extrapolation in RANS models”, expected Nov 2026.
- Advisor, Alberto Bianco: “Riduzione di modello per ODE parametriche mediante metodi a base ridotta”, expected Oct 2026.
- Advisor, Cosimo Becagli: “Physics informed neural networks”, expected Oct 2026.
- Co-advisor, Marcia Garcia Vinuela: “Neural Network extrapolation of VMS online Filters”, SISSA, Jul 2026.
- Advisor, Andrea Tataranni: “Physics Informed Neural Networks and Neural Tangent Kernel: preliminary results for parametric Optimal Control Problems”, Nov 2024.
- Co-advisor, Susanna Olivero: “Surrogate Models for Parametric PDEs via Graph-Informed Neural Networks”, Oct 2024.
- Co-advisor, Ilaria Anselmi: “Deep Learning e CFD: sviluppo di modelli per la previsione della visibilità in scenari d’incendio in galleria”, Oct 2024.
- Co-advisor, Carlotta Filippin: “Nonlinear reduced-order modeling with a Graph Convolutional Autoencoder for time-domain electromagnetics”, Oct 2024.
- Co-advisor, Fabio Zoccolan: “Stabilised reduced order methods for advection-diffusion optimal control problems with random inputs”, University of Trieste, Dec 2021.
- Co-advisor, Eleonora Donadini: “A Data-Driven Approach for Time-Dependent Optimal Control Problems by Dynamic Mode Decomposition”, University of Trieste, May 2021.
- Co-advisor, Giuseppe Carere: “Reduced Order Methods for Optimal Control Problems constrained by PDEs with random inputs and applications”, Korteweg-de Vries Institute, Jan 2019.

Seminars and Events

- Conference committees: COUPLED 2025, Scientific Machine Learning: Emerging Topics 2024.
- Minisymposium organizer: WCCM–ECCOMAS 2026, ECCOMAS YIC 2025, COUPLED 2025, SIAM UQ24, ECCOMAS24, WCCM24.
- Seminar organizer: NA-G-ROMs (2023–ongoing), Analysis Junior Seminars (2019–2021), SISSA Women in Mathematics 2021.

Other Tasks

- Research groups/projects: DISMA Numerical Analysis and Scientific Computing; external SISSA MathLab collaborator: ARIA, ERC AROMA-CFD, PRIN FaReX.
- Scientific groups: UMI, INDAM–GNCS, SIMAI, SIAM.
- Reviewer: Communications in Nonlinear Science and Simulation; SIAM Journal on Control and Optimization; Computer Methods in Applied Mechanics and Engineering; Journal of Computational Physics; International Journal for Numerical Methods in Engineering; International Journal for Numerical Methods in Fluids; Mathematics; Computers and Mathematics with Applications; Applied Mathematics and Computation; International Journal of Heat and Fluid Flow; Journal of Scientific Computing; Advances in Computational Mathematics; Advanced Modeling and Simulation in Engineering Sciences; Physica A; Frontiers in Applied Mathematics and Statistics; Proceedings in Applied Mathematics and Mechanics; International Journal of Computational Fluid Dynamics.

- President, SISSA SIAM Student Chapter (Oct 2020–Oct 2021); Vice-president (Oct 2019–Sep 2020).
- Educational volunteer, SISSA 4 SCHOOLS (2019–present); educational seminar, Pint of Science Festival.
- Internship: finite-element simulation of the quasi-geostrophic equation in the North Atlantic Ocean, OGS, 2016.